EpiHack Next-Gen

LLM Architecture & Prompts for One Health Participatory Surveillance

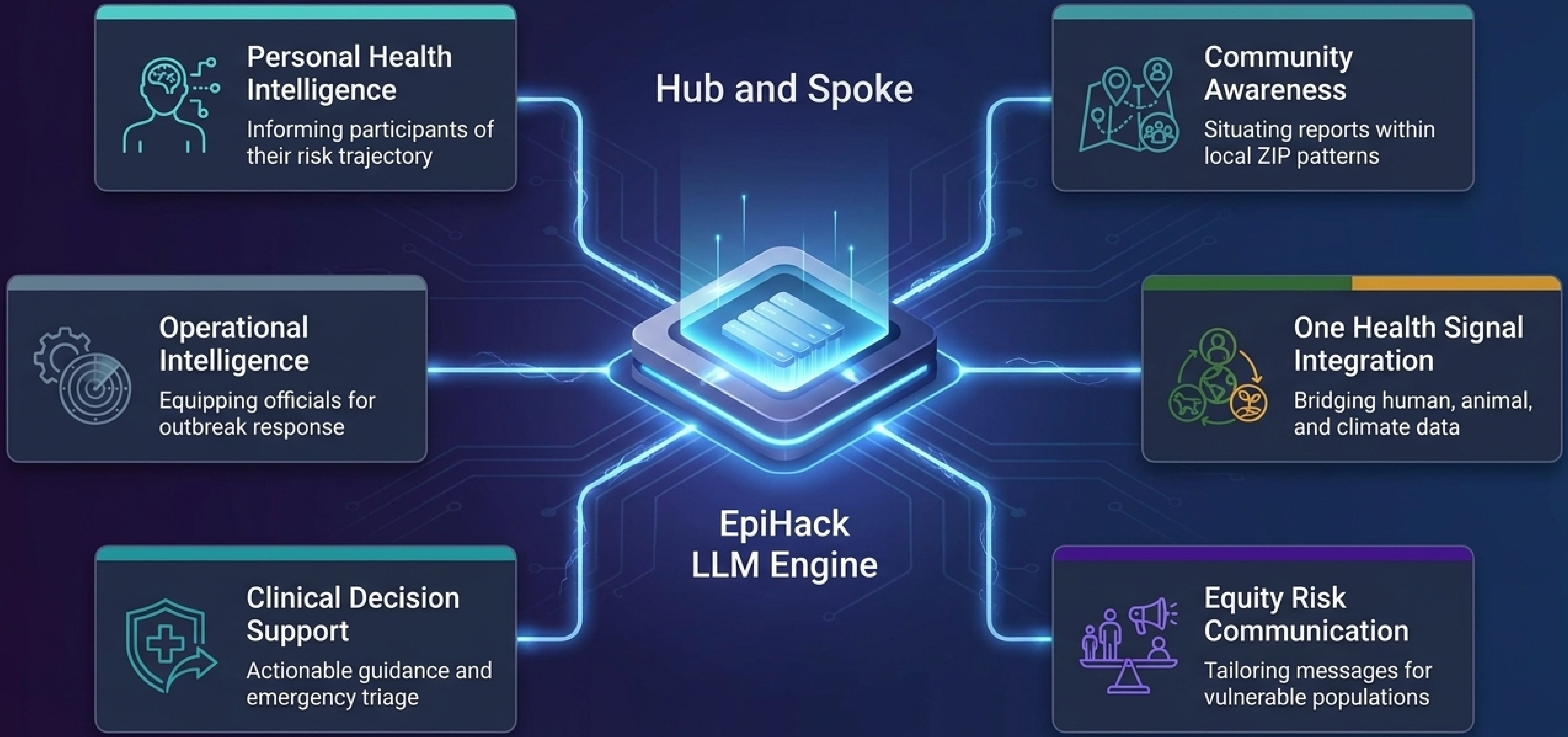
[Explainer] [Critical Summary] [Architectural Blueprint]


The Status Quo Prototype
Temporal Specificity
Symptom Granularity
Geographic Sensitivity
No One Health Data
Equity Blindspot
Missing Clinical Triage
Tribal Sovereignty Ignored
Open Loop / Survey Attrition
The 6-Pillar Next-Gen Framework
AI
NotebookLM

Diagnostic Comparison Matrix

	Status Quo Prototype	Next-Gen LLM Architecture
Scope	Human-only signals.	Integrated One Health (Human, Environmental, Zoonotic).
Equity	AZSVI ignored.	Demographics stratified by social vulnerability quintiles.
Clinical Support	Missing.	Automated Triage Escalation based on acuity.
Data Governance	Assumed universal access.	Explicit constraints for Tribal Data Sovereignty.

The EpiHack LLM Engine: The 6-Pillar Framework



Anatomy of an EpiHack LLM Prompt

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You are a One Health environmental health advisor.

The participant reported {symptoms} on {report_date}.
The NWS HeatRisk index for {postal_code} on that date was
 {heat_risk_level} (scale: 0–4).

In 3–4 sentences, explain the relationship between heat
 exposure and the reported symptoms, describe signs of
 heat illness requiring emergency care, and recommend
 protective behaviors.

Cite the heat risk level explicitly.
```

Priority: HIGH | Sources: NWS HeatRisk API

The Persona: Bounds system role to prevent hallucinations.

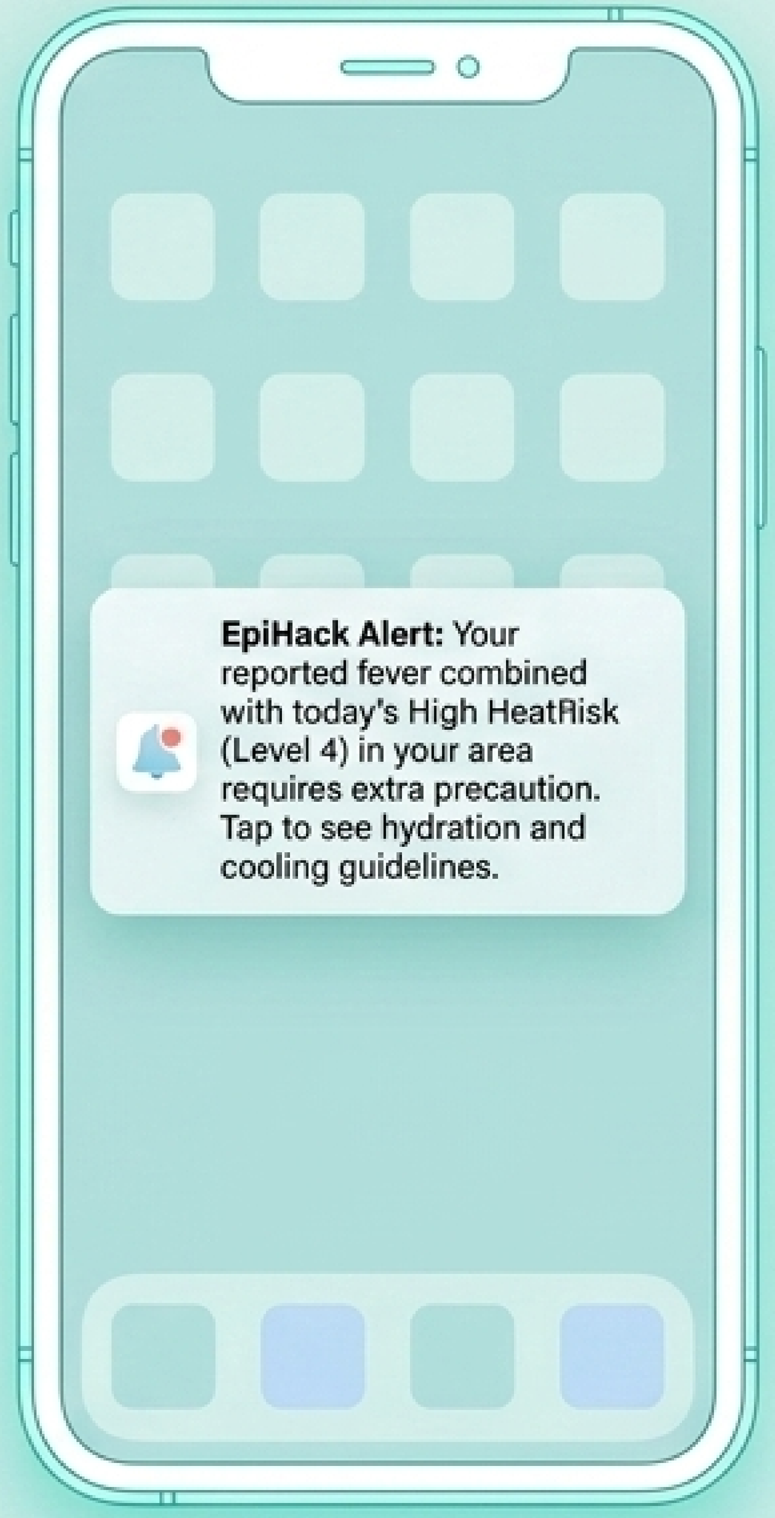
The Dynamic Ingest: Where the API injects live data.

The Guardrails: Ensures clinical safety and proper escalation.

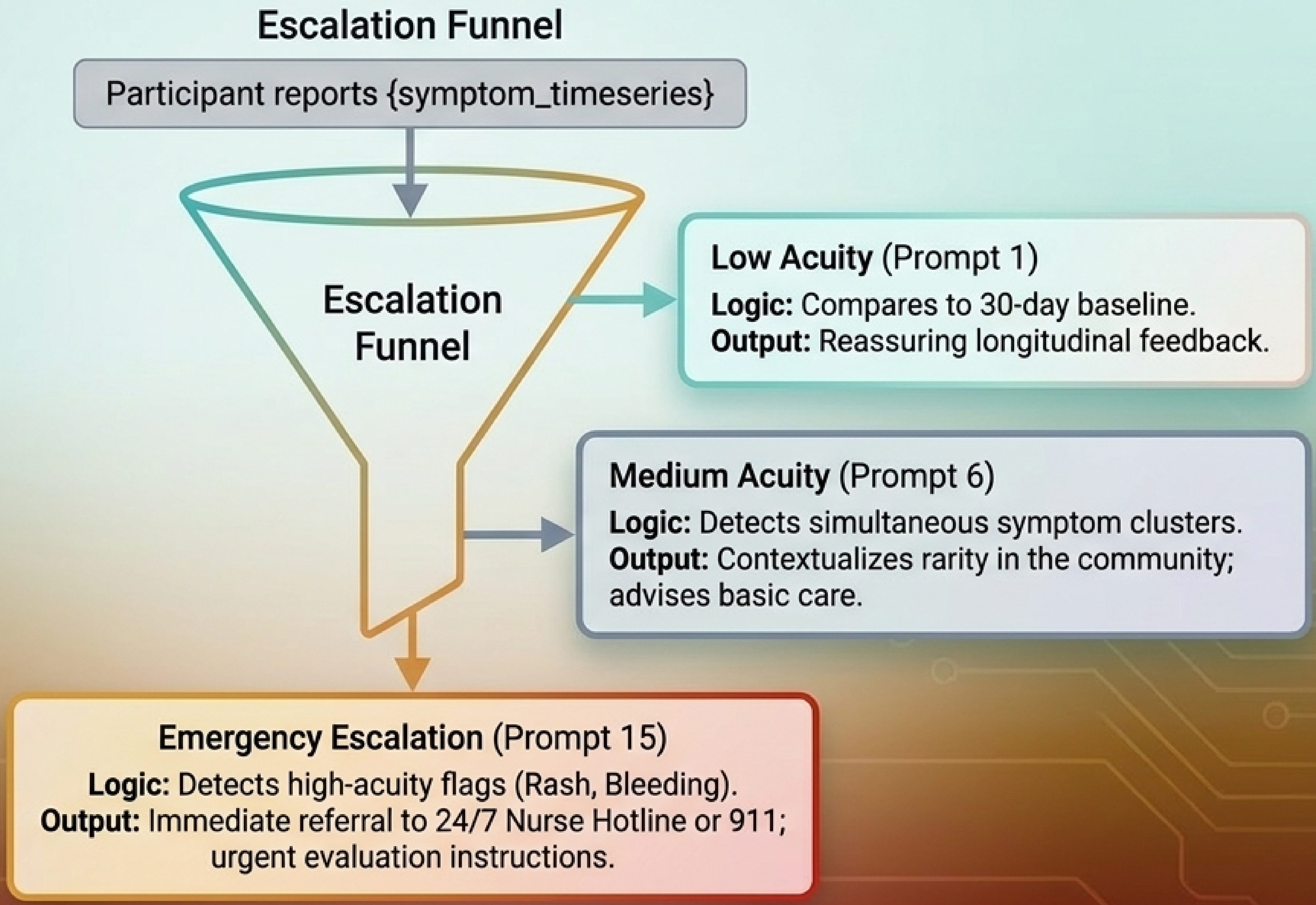
Track 1: The Participant Experience

Translating complex epidemiological data into personalized, engaging, and culturally responsive health intelligence.

Includes 20 participant-facing user stories optimized for engagement, safety, and health literacy.



Automated Clinical Triage Escalation



One Health in Action: Dynamic Contextualization

Environmental Stream

Raw Input:
Participant Fever/Muscle Aches
+ NWS HeatRisk Index 4

LLM Output (Prompt 4):
Warns of heat-related illness, cites specific HeatRisk level, and recommends immediate hydration and cooling.

One Health Integration

Zoonotic Stream

Raw Input:
Participant Fever+Rash
+ Postal Code 85001

LLM Output (Prompt 8):
Asks plain-language screening questions about recent livestock, wildlife, or tick exposure. Advises mentioning animal contact to physician.

Equity, Culture, and Trust

Culturally Responsive Health Literacy (Prompt 18)

- Translates epidemiological outputs to an 8th-grade reading level in Spanish.
- Uses culturally appropriate examples tailored for Arizona's Latino community.
- Strictly forbids literal word-for-word translation, prioritizing true comprehension.

Tribal Data Sovereignty (Prompt 19)

- Acknowledges and respects data governance for Arizona's 22 recognized tribes.
- Drafts messages referencing specific tribal health departments.
- System strictly forbidden from sharing individual or aggregate data without explicit tribal governance authorization.

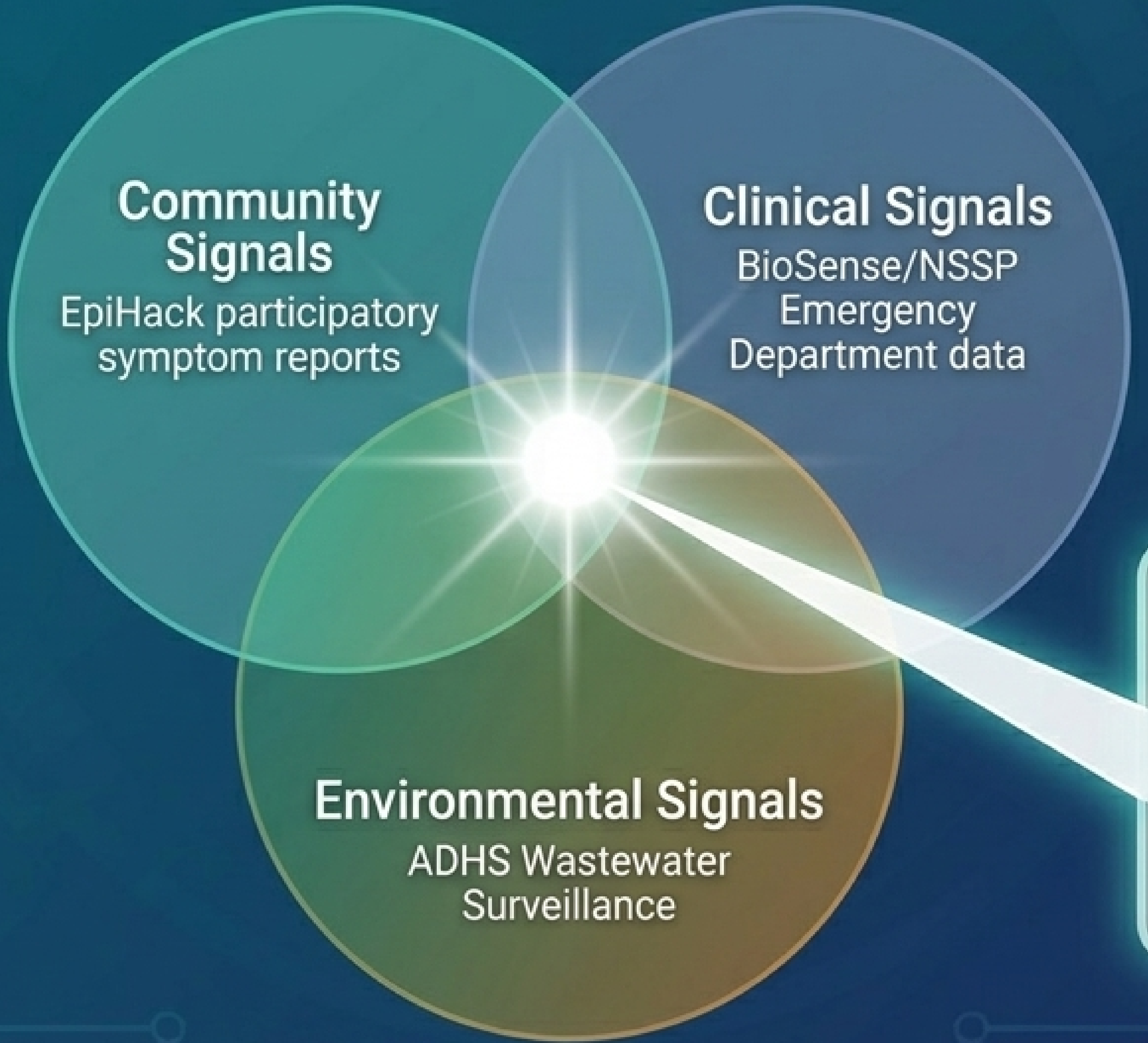
Track 2: Public Health Operational Intelligence

Synthesizing noise into actionable signals for institutional decision-makers.

Includes 20 official-facing prompts designed for cluster detection, resource allocation, and automated hypothesis generation.

Multi-Source Triangulation (Prompt 5)

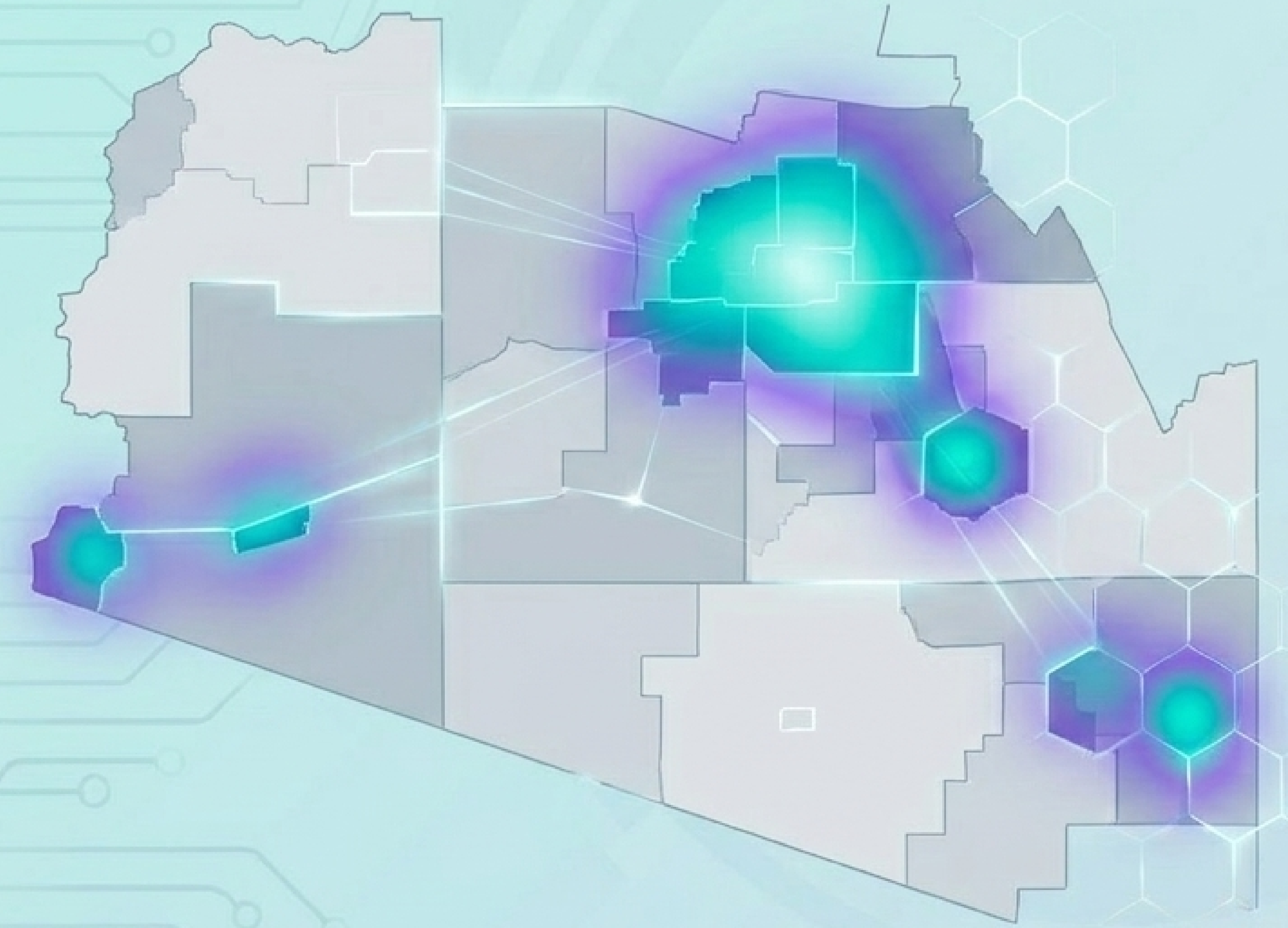
Cross-referencing data streams to confirm outbreaks and reduce false alarms.



Tier 1 Signal Confidence

When 2+ data sources are concordant, the LLM integrates them to output a structured decision table, assigning a high-confidence signal and automated response level.

Cluster Detection & Resource Allocation



Automated Deployment Logic (Prompts 2 & 12)

Detect Multi-Symptom Clusters
Flags postal areas with 3+ elevated symptoms showing 'up' or 'new' signals.

↓

Apply Equity Stratifier
Multiplies burden by AZSVI (Social Vulnerability Index) to generate a "Composite Need Score".

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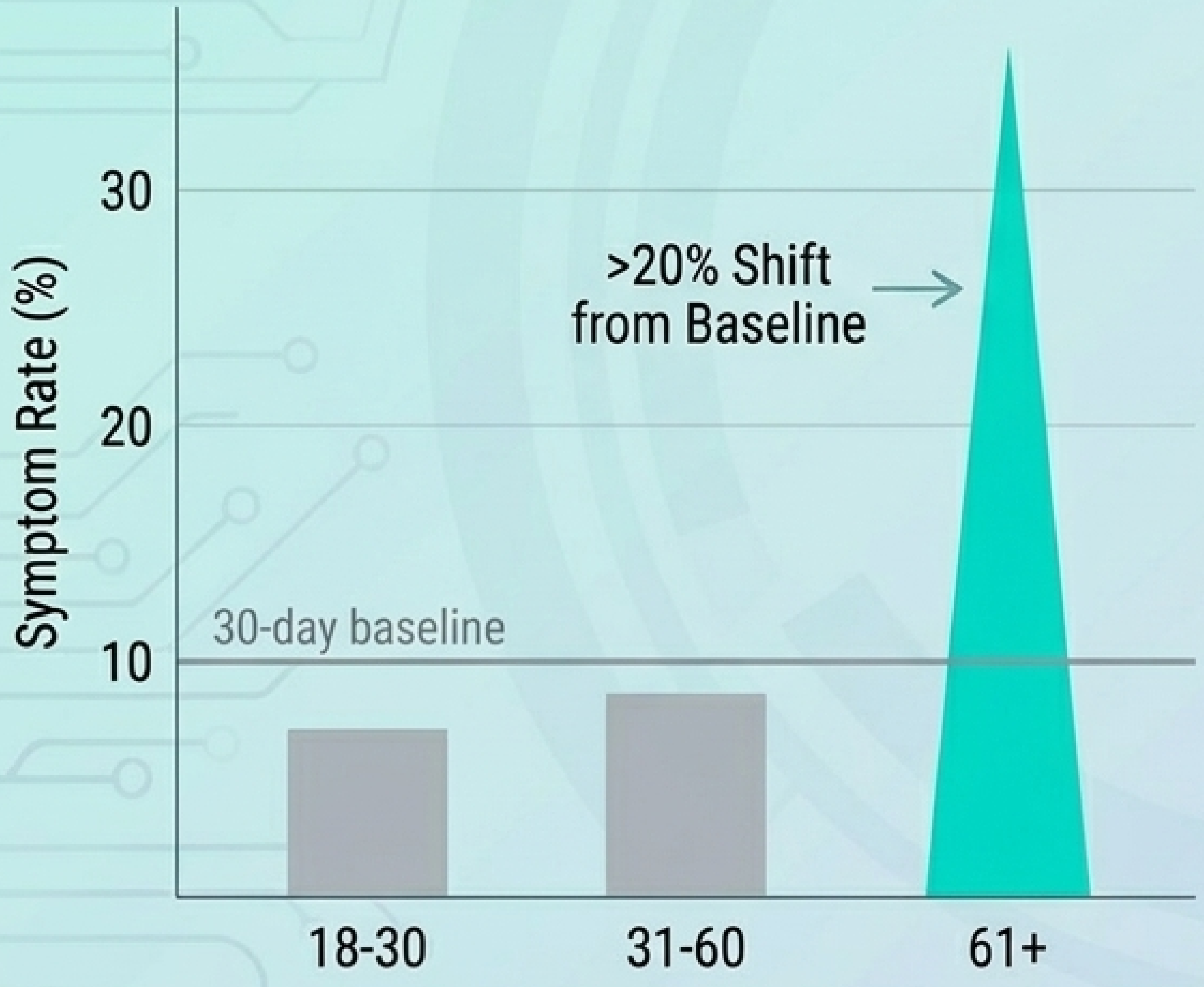
Logistical Output
Generates automated recommendations to deploy mobile testing units, PPE, and field workers to the highest-need postal areas.

Demographic Anomalies & Hypothesis Generation

Shift Detection (Prompt 7)



Automated Research Assistant (Prompt 16)

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[Hypothesis]:
Observed symptom shift in the 61+ demographic
is driven by heat-exacerbated respiratory
distress rather than a novel pathogen.

[Proposed Study Design]:
Case-control study matching 61+ reporters
against asymptomatic peers in identical AZSVI
quintiles.

[Required Data Sources]:
EpiHack symptom timeseries, NWS HeatRisk API,
ADHS ED discharge data.

[Feasibility Score]: 4/5
```
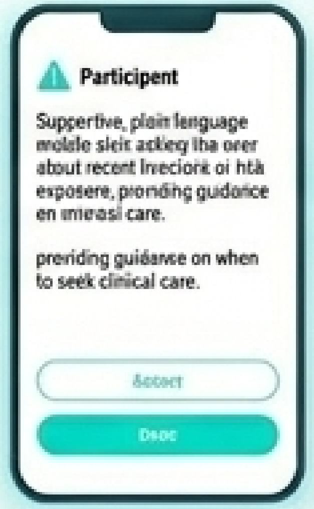

The One Health Data Engine: Dynamic Audience Translation



Participant Output

Supportive, plain-language mobile alert asking the user about recent livestock or tick exposure, providing guidance on when to seek clinical care.

RAW DATA INPUT

Fever + Rash cluster detected in ZIP 85001

Official Output

Technical situation report correlating the human cluster with ArboNET vector data, calculating a Composite Need Score, and recommending field resource allocation.

Key Takeaway: One unified data engine, dynamically contextualized for two entirely distinct audiences.

The Participatory Feedback Loop

Solving survey attrition by transforming one-way surveillance into a trust-building ecosystem.

Participant reports a symptom to the platform (Track 1).

Public Health Officials detect a cluster and deploy resources (Track 2).

LLM ingests public health action and generates “Progress Feedback” (Prompt 16).

Participant receives a personalized notification.